

with Program Specific Outcomes(PSO)	CO3								
	CO4								
	CO5								
	CO6								
Course Content	Unit-1: Fundamentals of AI and Machine Learning								
	1.1 Concepts of Machine Learning 1.1.1 Understanding Machine Learning 1.1.2 Benefits of Machine Learning 1.1.3 Difference between AI, ML, and Deep Learning 1.2 Machine Learning Life Cycle 1.2.1 Stages of ML Life Cycle 1.2.2 Problem Definition to Model Monitoring 1.3 Exploratory Data Analysis (EDA) 1.3.1 Types of EDA 1.3.2 Univariate Analysis 1.3.3 Bivariate Analysis 1.3.4 Multivariate Analysis 1.3.5 Handling Missing Data 1.3.6 Detecting and Handling Outliers 1.4 Understanding the Data 1.4.1 Quantitative Data 1.4.1.1 Discrete Data 1.4.1.2 Continuous Data 1.4.2 Qualitative Data 1.4.2.1 Nominal Data 1.4.2.2 Ordinal Data 1.5 Spread of Data 1.5.1 Normal Distribution 1.5.2 Skewed Distribution 1.5.3 Skewness 1.5.4 Kurtosis								
	Unit-2: Automate EDA (Exploratory Data Analysis)								
	2.1 Python Libraries to Automate EDA 2.1.1 Pandas, NumPy, D-Tale 2.1.2 Pandas Profiling 2.1.3 Sweetviz AutoViz 2.2 Regression Fundamentals 2.2.1 Characteristics of Regression 2.2.2 Dependent and Independent Variables 2.2.3 Covariance 2.2.4 Correlation 2.3 Simple Linear Regression (SLR) 2.3.1 Understanding SLR 2.3.1.1 Fitting a Straight Line (y = mx + b) 2.3.1.2 Identifying Dependent and Independent Variables 2.3.2 Assumptions of SLR 2.3.2.1 Linearity 2.3.2.2 Independence of Errors 2.3.2.3 Homoscedasticity 2.3.2.4 Normality of Errors 2.3.3 Goodness of Fit 2.3.3.1 R² and Adjusted R² 2.3.3.2 Significance of R² Value 2.3.3.3 Residual Sum of Squares (RSS)								

	Unit-3: Understanding Machine Learning 3.1 Types of Machine Learning 3.1.1 Supervised Learning 3.1.2 Unsupervised Learning 3.1.3 Applications of Machine Learning 3.2 Overview of Supervised Learning 3.2.1 Concepts of Supervised Learning 3.2.2 Classification vs. Regression 3.3 Basic Terminologies 3.3.1 Dataset 3.3.2 Features and Labels 3.3.3 Training, Testing, and Validation Data 3.3.4 Overfitting and Underfitting 3.4 Loss Functions 3.4.1 Mean Squared Error (MSE) 3.4.1.1 Definition of MSE 3.4.1.2 Computation and Properties 3.4.2 Mean Absolute Error (MAE) 3.4.2.1 Definition of MAE 3.4.2.2 Computation and Properties Unit-4: Supervised Learning Algorithms 4.1 Linear Regression (Review and Model Evaluation) 4.1.1 Model Fitting using Ordinary Least Squares (OLS) 4.1.2 Model Evaluation Techniques 4.1.2.1 R-squared 4.1.2.2 Mean Squared Error 4.1.2.3 Root Mean Squared Error (RMSE) 4.2 Logistic Regression 4.2.1 Sigmoid Function and Binary Classification 4.2.2 Cost Function and Gradient Descent 4.2.3 Multiclass Classification using Logistic Regression 4.2.3.1 One-vs-Rest Strategy 4.2.3.2 Softmax Function 4.3 Model Performance and Generalization 4.3.1 Overfitting 4.3.2 Underfitting 4.3.3 Avoiding Overfitting/Underfitting 4.3.3.1 Regularization (L1 and L2) 4.3.3.2 Cross-Validation 4.3.4 Evaluation Metrics 4.3.4.1 Accuracy 4.3.4.2 Precision 4.3.4.3 Recall 4.3.4.4 F1 Score
Reference Books	1) Machine Learning, T.V. Vijay Kumar, BS Publications, ISBN:9789381075614 2) Fundamentals of Machine Learning for Predictive Data Analytics, K.P. Soman, R. Loganathan, V. Ajay, PHI Learning Pvt. Ltd., ISBN: 9788120350786 3) Programming in Python, Dr. Rakesh K. Jain, University Science Press (Laxmi Publications), 9789386768016 4) Data Analytics with Python, Suresh Samudrala, V. Rajaraman, PHI Learning Pvt. Ltd., 9789353062026 5) Introduction to Machine Learning, Alok Sharma, Narosa Publishing House, 9788184874535 6) Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer, 9780387310732

	7) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Aurélien Géron, O'Reilly Media, 9781098125974 8) Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, MIT Press, 9780262018029 9) Python Machine Learning, Sebastian Raschka, Vahid Mirjalili, Packt Publishing, 9781800567708 10) The Elements of Statistical Learning, Trevor Hastie, Robert Tibshirani, Jerome Friedman, Springer, 9780387848570
Teaching Methodology	Class Work, Discussion, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.